

Segmentation of brain from computed tomography head images

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Abstract - An algorithm to determine the human brain (gray matter (GM) and white matter (WM)) from computed tomography (CT) head volumes with large slice thickness is proposed based on thresholding and brain mask propagation. Firstly, a 2D reference image is chosen to represent the intensity characteristics of the original 3D data set. Secondly, the region of interest of the reference image is determined as the space enclosed by the skull. Fuzzy C-means clustering is employed to determine the threshold for head mask and the low threshold for brain segmentation. The high threshold is calculated as the weighted intensity average of the boundary pixels between bones and GM/WM. Based on the low and high thresholds, the CT volume is binarized, followed by finding the brain candidates through distance criterion. Finally the brain is identified through brain mask propagation using the spatial relationship of neighboring axial slices. The algorithm has been validated against one non-enhanced CT and one enhanced CT volume with pathology.

1. INTRODUCTION

With the advent of multidetector computed tomography (CT) and fast scan times, CT has become the centerpiece for cranial imaging. It is the examination of choice for investigating stroke, intracranial hemorrhage, trauma and degenerative diseases (<http://www.neurorad.com/ct02.htm>). CT has several advantages over magnetic resonance imaging (MRI). These include short imaging times, widespread availability, ease of access, optimal detection of calcification and hemorrhage and excellent resolution of bony detail. CT is also valuable in patients who cannot have MRI due to implanted biomedical devices or ferromagnetic foreign material.

Limitations to CT scanning of the head include: artifacts due to beam hardening and spiral off-centre, radiation burden on the patient, and low brain tissue contrast.

In spite of these drawbacks, CT continues to be the main stay in cranial pathology management tools.

Unlike the rich literature in brain segmentation from MRI data, research on segmenting brain from CT images is sparse. Maksimovic et al. [1] used active contour to find lesion and ventricles of patients with acute head trauma

with manual drawing of initial contour. Deleo et al. [2] proposed a semi-automatic method. Users were requested to manually select representative points of cerebrospinal fluid (CSF), gray matter (GM), and white matter (WM) in the region superior to the third ventricle to avoid beam hardening. Thresholds were calculated based on the manual specification of the representative CSF, GM and WM. Its drawbacks include: manual specification is tedious and error prone without training, beam hardening cannot be handled, not the whole brain is covered for categorization, and spatial information is not exploited to deal with tissues having overlapping intensity. Ruttimann et al. [3] proposed to use maximum between-class variance criteria for differentiating hard and soft tissues, and CSF was segmented using a local thresholding technique based on maximum-entropy principle. The processing was limited to selected axial slices. Soltanian-Zadeh and Windham [4] proposed to find brain contours in a semi-automatic way: manually specify the thresholds at different regions to binarize CT slices, use edge tracking to find contours, use multi-resolution to resolve broken contours, and specify seed points to pick up the desired contour. The vast amount of user intervention is its major drawback. In this paper we propose to extract the brain from CT data based on thresholding and brain mask propagation. Two thresholds are automatically determined: the high threshold is to exclude the high intensity bones, while the low threshold is to exclude air and CSF. Brain mask propagation is to make use of the spatial relevance of brain tissues in neighboring slices to exclude non-brain tissues with similar intensities.

2. METHOD

Brain consists of GM and WM. In CT brain images, bones have the highest intensity, followed by GM, WM, CSF, and air. Non-brain tissues like various sinuses and muscles may have similar intensities to GM or WM. Due to the invasive nature of CT imaging, the slice thickness is normally large (≥ 5 mm) to decrease subject's exposure to radiation. The implication of the

large slice thickness is that neighboring axial slices have some relationship, but one cannot assume that the brain tissues as a whole will form the largest connected component as done in the case of MRI with small slice thickness.

The coordinate system (xyz) of a volumetric dataset is represented according to the standard radiological convention: x runs from subject's right to left, y from anterior to posterior, and z from superior to inferior. The intensity of a voxel (x, y, z) is denoted as $g(x, y, z)$. An axial slice consists of those voxels with z being a constant. Fig. 1 shows the flow chart of the proposed method.

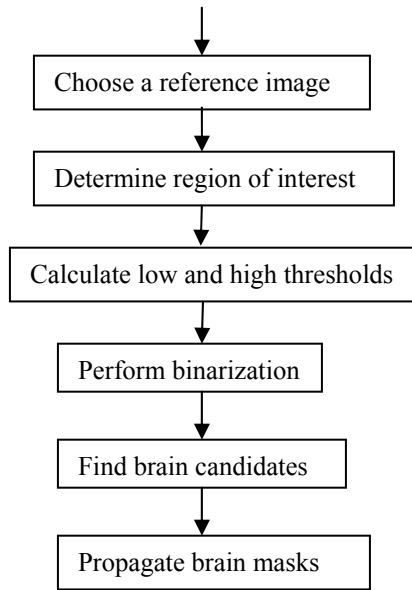


Fig. 1: flow chart for segmenting the brain from CT head volumes.

Firstly, a two-dimensional (2D) reference image is chosen to well represent the intensity characteristics of the original three-dimensional (3D) data. Then a region of interest (ROI) is determined where the object (GM and WM) is present. Within the ROI the low and high thresholds are determined based on the spatial and intensity relationship between the brain and non-brain tissues. Then the head volume is binarized using the two thresholds. Connected components and distance criteria are combined to get the brain candidates, which are further refined through propagation of brain masks to find the final brain mask.

2.1 Choose a reference image $g(x, y, z_0)$

The reference image is a 2D image from the 3D volumetric CT data set. The reference image should have the following characteristics: it has WM, GM, CSF, air, and skull tissues present; it is easily extracted from the volume anatomically; the proportion of GM and WM within the skull should only vary in a narrow range. A possible reference image is the

axial slice passing through the anterior and posterior commissures. In practice, this reference image can be approximated by an axial slice with the third ventricle present and without eyes, as shown in Fig. 2a. The axial slice number is denoted as z_0 , and the reference slice is denoted as $g(x, y, z_0)$.

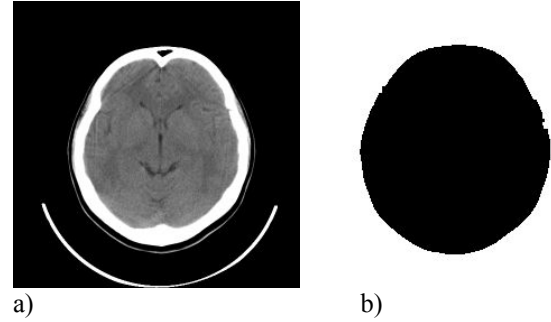


Fig. 2: the reference image and the corresponding ROI.

2.2 Determine region of interest

The ROI of the reference image is the space enclosed by the skull and is called head mask subsequently. It is determined as follows:

- 1) Find the threshold to binarize the reference slice. From the intensity histogram of the volume $g(x, y, z)$ classify the intensity into 4 clusters using fuzzy C-means (FCM) clustering (www.elet.polimi.it/upload/matteucc/Clustering/tutorial_html/cmeans.html), with the first cluster having the smallest intensity. The maximum intensity of the first cluster plus 5 is denoted as backG.
- 2) Binarize $g(x, y, z_0)$ to get the initial head mask $headM(x, y, z_0)$: $headM(x, y, z_0)$ is set to 0 (background) if $g(x, y, z_0)$ is smaller than backG, and 1 (foreground) otherwise.
- 3) Find the largest foreground connected component and set other foreground components of $headM(x, y, z_0)$ to background.
- 4) Fill the holes (changing 0 to 1) within the largest connected component to get the final head mask $headM(x, y, z_0)$.

Fig. 2b shows the ROI of Fig. 2a.

2.3 Calculate low and high thresholds

The low threshold is to exclude air and CSF from the brain. It is determined by the following sub-steps:

- 1) From the intensity histogram of the reference image in the head mask $headM(x, y, z_0)$, classify the intensity into 4 clusters corresponding to air and CSF, WM, GM, and bone. Denote the smallest intensity of the fourth cluster as minBone.
- 2) The low threshold is calculated as: $lowThresh = meanC_1 + \alpha_1 * sdC_1$, where $meanC_1$ and sdC_1 are the mean and standard deviation of cluster 1, while

α_1 is a constant in the range of 0 and 3. When it is required to have less brain tissues classified as non-brain tissues, α_1 should be small, say, smaller than 1; otherwise, if one is more interested in separating brain from non-brain tissues, α_1 should be big, say bigger than 2.

The high threshold serves to exclude bone that is brighter than both GM and WM. It looks that the bright bone is spatially adjacent to GM and WM, though physically bones are not exactly adjacent to GM or WM. The high threshold is determined from pairs of boundary pixels between bones and WM/GM in the reference image as follows.

- 1) Within the head mask $headM(x, y, z_0)$ of the reference image find all pairs of pixels satisfying: a) the pair is 8-connected; c) the intensity of one pixel is not smaller than $minBone$ (a bone pixel) and the intensity of the other is smaller than $minBone$ but bigger than $lowThresh$ (a GM or WM pixel).
- 2) For all the pairs of pixels found in 1), calculate the intensity average of pixels with intensities not smaller than $minBone$ and denote it as $brightAvg$. Similarly, calculate the intensity average of pixels with intensities smaller than $minBone$ and denote it as $darkAvg$.
- 3) The high threshold is determined by $highThresh = \alpha * brightAvg + (1 - \alpha) * darkAvg$. Here α is a constant in the range of 0 and 1. If the cost to exclude brain tissue is bigger than the cost to include non-brain tissue in the segmentation, α should be bigger than 0.5. If both costs are equally important or one wants to have minimum classification error then α should be 0.5.

2.4 Perform binarization

Binarization is performed on the original CT volume $g(x, y, z)$ to get the binary mask $binM(x, y, z)$

$$binM(x, y, z) = \begin{cases} 1, & lowThresh \leq g(x, y, z) \leq highThresh \\ 0, & otherwise \end{cases}$$

Fig. 3a shows the resulted binary mask of Fig. 2a.

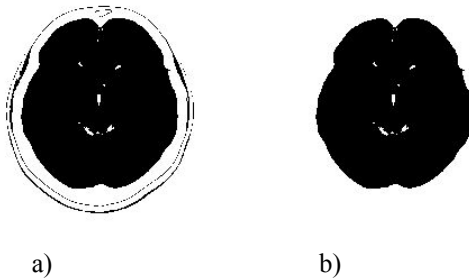


Fig. 3: binarized and brain candidate of the reference image shown in Fig. 2a.

2.5 Find brain candidates

For all axial slices, their head masks are found as described in 2.2. For any axial slice z , find all the foreground

connected components of $binM(x, y, z)$. When the smallest distance to the background pixels of the head mask of the axial slice is bigger than a constant (say, 10 mm), the foreground component is taken as the brain candidate; otherwise, it is set to background (skull). Figs. 3b and 4 show the brain candidates of the reference image and an axial slice inferior to the reference image, respectively. Note in Fig. 4, there are non-brain regions left to be removed in the next stage.

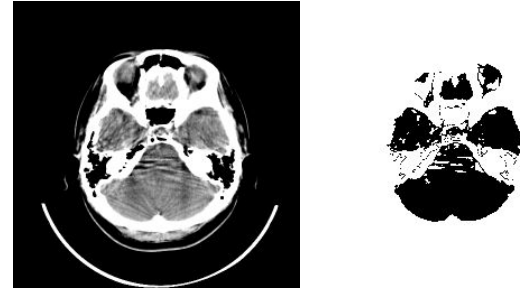


Fig. 4: an axial slice inferior to the reference slice and its brain candidates after applying distance criterion.

2.6 Propagate brain masks

The non-brain regions can be removed through brain mask propagation. Specifically, check all the brain candidates with $z > z_0$ consecutively starting from slice z_0+1 . In slice z_0+1 , check all the foreground components the following way:

- 1) For a foreground connected component at slice z_0+1 , suppose the number of brain candidate pixels is N , and the connected component is a point set $\bigcup_i (x_i, y_i, z_0 + 1)$. For all (x_i, y_i) , count the number of brain candidate pixels (x_i, y_i, z_0) at slice z_0 and denote it as N_1 .
- 2) If N_1 is smaller than $\beta * N$, it indicates that the connected component at slice z_0+1 is very different from the brain contents at axial slice z_0 , this foreground connected component at slice z_0+1 is turned to background. Here β is a constant in the range of 0 to 1, typically it takes the value of 0.5.

After all the foreground connected components in slice z_0+1 are checked, proceed to slice z_0+2 . This process continues until all the axial slices with z bigger than z_0 have been checked. The left brain candidates are the brain. Fig. 5 shows the final brain after the brain propagation of the axial slice shown in Fig. 4a.

3. RESULTS

The algorithm was implemented in C++ on Pentium 4 (2.6 GHz CPU). It was quantitatively validated against 1 non-enhanced CT data set (256x256x19 voxels, 1.0x1.0x5.0 mm³ voxel size) and qualitatively tested on an enhanced CT data set (pathology and vessels are enhanced) with pathology (the same number of voxels

and voxel size). The brain was extracted within 90 seconds. The ground-truth brain contour was drawn by a neuroanatomy expert (AA) using the image editor Adobe Phtshop on all the axial slices.

False negative (FN) and false positive (FP) were used to quantify the accuracy of segmentation. The numerators of FP and FN are the number of brain voxels present in the segmented brain but not in the ground truth, and the number of brain voxels present in the ground truth but not in the segmented brain respectively, with the number of brain voxels in the ground truth as the same denominator. For the normal CT data set, the yielded FN and FP are 2.03% and 3.12%, respectively. The misclassified voxels are mainly due to the partial volume effect of ventricles.

For the enhanced CT data set with pathology, the segmentation is visually checked with satisfaction. Fig. 6 shows the segmented brain of an axial slice.

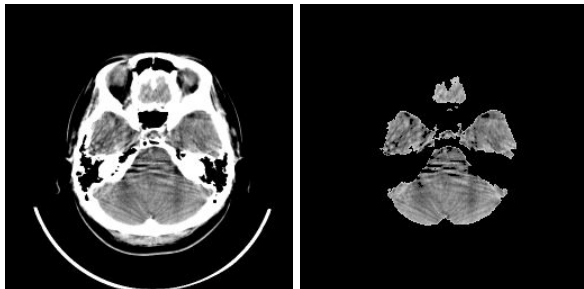


Fig. 5: the final brain after brain mask propagation.

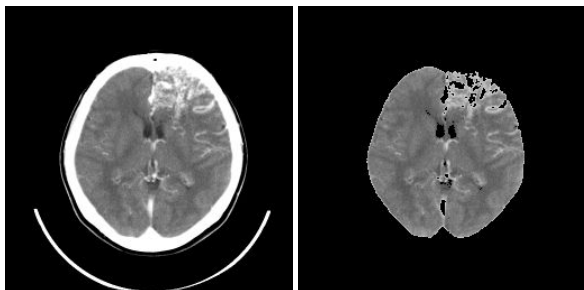


Fig. 6: segmented brain for the enhanced CT with pathology.

4. DISCUSSION AND CONCLUSION

Segmenting brain from clinically available CT data with large slice thickness is of importance as it can help to interpret the anatomical structures, to identify pathology and help registration. An algorithm based on thresholding and brain mask propagation is proposed Firstly, a 2D reference image is chosen to represent the intensity characteristics of the original 3D data set. This reference image can be the axial slice passing through the anterior and posterior commissures, or approximated by an axial slice with the third ventricle present and without the orbits. The appropriate selection of the reference image simplifies thresholding from 3D data to 2D images. Secondly, the ROI

of the reference image is determined as the space enclosed by the skull. FCM clustering is employed to determine the threshold for head mask and the low threshold for brain segmentation. FCM is advantageous over curve fitting [5] as it does not assume the noise model and is applicable even in the presence of heavy noise and artifacts. The high threshold is calculated as the weighted intensity average of the boundary pixels between bones and WM/GM. Based on the low and high thresholds, the CT volume is binarized, followed by finding the brain candidates through distance criterion. Finally the brain is identified through brain mask propagation using the spatial relationship of neighboring axial slices.

Non-brain tissues with similar intensities (like muscles and sinuses) to WM or GM have been effectively removed through distance criterion and brain mask propagation.

In the steps to determine the low and high thresholds, we introduced two parameters α_1 and α . It is obvious that increasing α_1 and/or decreasing α will decrease FP but increase FN. Similarly, decreasing α_1 and/or increasing α will decrease FN but increase FP. Thus the two parameters (α_1 and α) provide a mechanism to balance the FN and FP rates for real applications, which is a unique feature of our methods.

Test against two CT data sets shows that the beam hardening effect has been tackled well with the low and high thresholds determined. More data sets need to be investigated, though.

References

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